DAURES GREEN HYDROGEN VILLAGE

University of Stuttgart
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4.3 Implementing TIMES Namibia (TINa) Energy System Model

OVERVIEW

This factsheet covers the steps used for the development of the TIMES Namibia (TINa) energy system model. The general TIMES energy system modelling frameworks are discussed in Factsheet 4.2; Namibia's energy profile overview, also crucial to understanding TINa model is discussed in Factsheet 4.1. The model explores the impacts of hydrogen and derivatives on Namibia's energy system, and how these play a role in achievement of Namibia's policy objectives.

Based on the TIMES modeling framework, TINa is a full sector model which covers the full supply chain from extraction or harnessing of primary energy carriers to energy service demands as required by the whole economy. It includes a detailed representation of key components of the energy system i.e. energy imports, domestic supply and electricity generation, representation of energy-intensive industries transport sector by sub-categories, residential energy consumption for different living conditions and electricity access. Besides energy-related GHG emissions, TINa also includes full coverage of process GHG emissions.

TINa OBJECTIVES AND POLICY OBJECTIVES

Research (TINa) objectives are based on Namibia's current policy (energy and climate) objectives; to explore the role of hydrogen on the energy system.

Particular strengths of TINa:

- Detailed representations of energy technologies and their linkages across sectors;
- Consideration of inter-dependencies in the energy system;
- Analysis of competition and substitution effects between technologies;
- Detailed representation of results: energy flows, new capacities, emissions, costs.

TINa is a useful tool to:

- Support decision-making and explore long-term options;
- Assess competition between energy carriers and end-uses;
- Determine how best to achieve specific energy and climate-related targets;
- Assess associated costs and emissions of different investment decisions;
- Analyze options for hydrogen production, direct use or export.
- Analyze the impact of H2 production on the existing infrastructure (generation + electricity grid + gas grid etc.) and subsequent changes in the energy system.

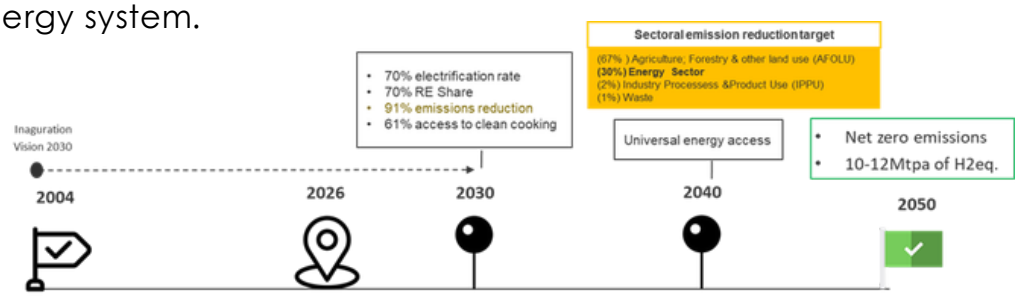

Figure 1. Energy and environmental objectives of Namibia

Namibia's Energy and environment Policy Objectives:

- Achievement of universal energy access, and energy security.
- Decarbonization: Achievement of 70% renewable energy share, reduction of GHG emissions by 21.9 MtCO2e (91%) by 2030; Achievement of net-zero emissions in 2050.

Research objectives:

The overall objective is to explore the role of hydrogen and derivatives in the Namibia 's energy system and support decision making with regards to:

- Potential for hydrogen production and pathways for use in Namibia, Southern Africa, Germany and other export regions;
- Competition between hydrogen and other energy carriers;
- Determine how best to achieve energy and climate-related targets.

TINa MODELLING STEPS

The five (5) steps used to develop the TINa model are as follows:

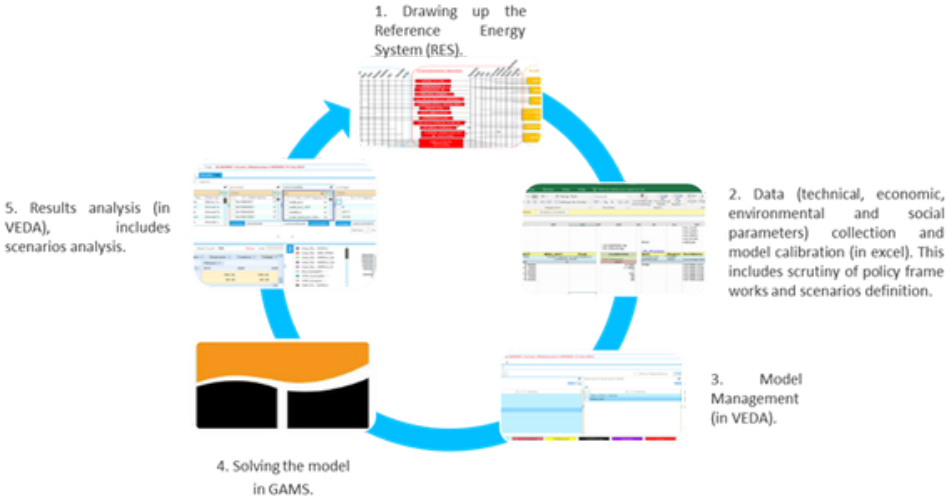

Figure 2. Steps to develop TINa model

TINa GENERAL STRUCTURE

TINa is a full sector model with Residential, Commercial and public service, Agriculture, Transport, Industry (including mining and steel) and the supply sectors. The model horizon: 2018 (base year) to 2050. Data inputs to the model includes technology, economic environment and social parameters for all processes and energy carriers.

The import, export region regions and user constraints as indicated in figure 3 below. The objective function as an output of the model is the discounted system cost required to meet the energy service demand under the constraints.

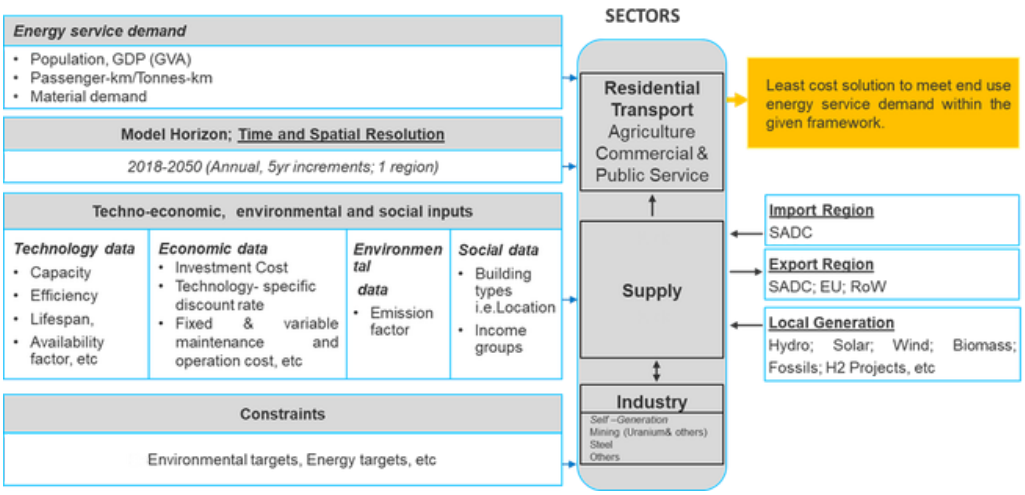

Figure 3. TINa general structure

TINa REFERENCE ENERGY SYSTEM (RES)

TINa's RES includes domestic supply and import of primary energy carriers, final energy carriers, conversion and demand technologies, end use energy demand services and materials for different demand sectors as summarized in the figure 4 below.

The RES serves as a map of the model inputs, outputs and interdependencies of various sectors, fuels and technologies.

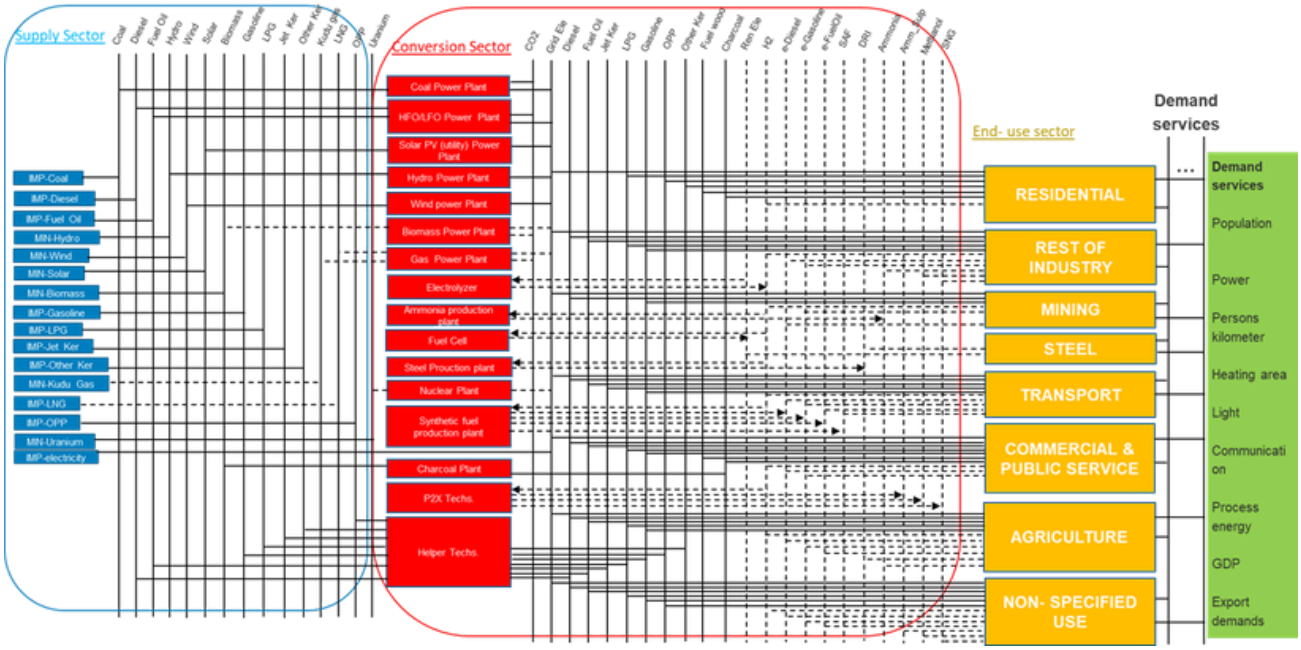

Figure 4. TINa reference energy system

SCENARIO ANALYSIS AND MODEL APPLICATIONS

Scenario analysis and applications of the TINa model are discussed in the factsheet 4.4 TINa Model Results and Applications.

CONCLUSIONS & SUMMARY

- TINa model explores the impacts of hydrogen and derivatives on Namibia's energy system.
- TINa includes disaggregated sectors, spatial and time resolution to achieve a high level of details, hence detailed results.
- TINa can be applied to support decision-making and explore long-term options, not limited to energy and climate related targets.

REFERENCES

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[2] https://www.nampower.com.na/Page.aspx?p=178
[3] R. Loulou et al., Energy Technology Systems Analysis Programme: Documentation for the TIMES Model PART I. 2016.

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Further Fact Sheets covering topics from Project Descriptions, Use Cases, Techno-Economic analysis, Social Indicators, Energy System Analysis, Macro-Economic modelling, and more are available:

https://github.com/IER-Hy4Daures/Fact-Sheets